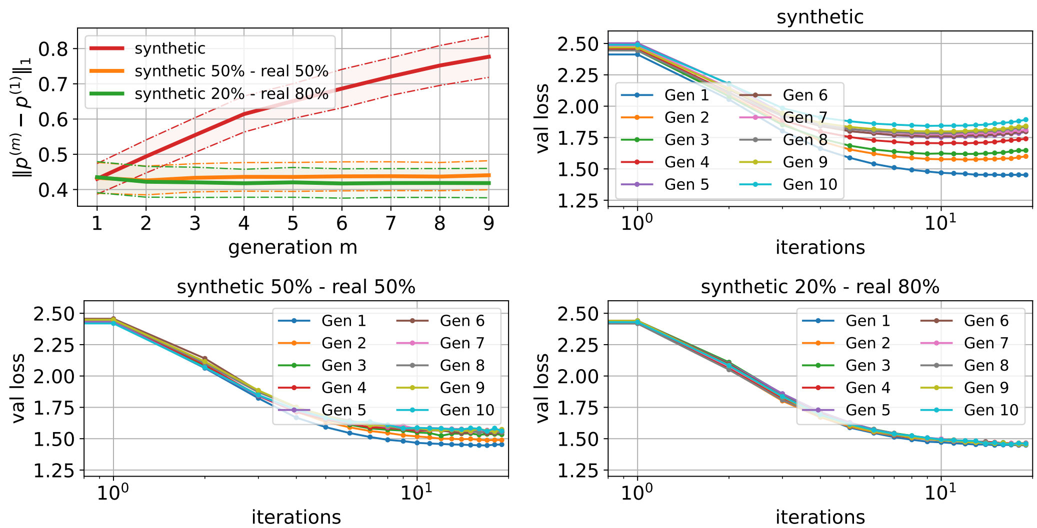

Figure 5: Experiments with a GPT2-type generative model. The top left plot depicts the deviation $\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1$ varying the generation model m for synthetic data and a mixture of real and synthetic data. The three other plots show the behavior of the validation loss over generations. Essentially, training solely on synthetic data causes model collapse and affects the usual scaling laws (Dohmatob et al., 2024a).

high-dimensional Gaussian vectors instead of canonical vectors, thereby introducing the embedding dimension as a parameter controlling model complexity.

4.2 Additional Experiments with the Statistical Model

To further investigate and demonstrate the generality of our theoretical findings, we present empirical results on two more scenarios that better represent recursive training in real-world settings, as follows:

- *Most Recent Models:* Each generation $p^{(m)}$ is trained on synthetic data from the most recent K models for a fixed window size K. More precisely, for some fixed $n \in \mathbb{N}$ we let $n_t^{(m)} = \lfloor n/K \rfloor \cdot \mathbb{1}\{\max(0, m-K) \leq t \leq m-1\}$ for all $m \geq 1$. When $K = 1$, this case degenerates to the *Fully Synthetic* setting.
- *Randomly Sampled Data:* Each generation $p^{(m)}$ is trained on a mixture of synthetic data from possibly all the previous models and the real data. More precisely, the m-th generation model is trained on $n = n_0^{(m)} + \cdots + n_{m-1}^{(m)}$ samples with

$$n_t^{(m)} := \sum_{i=1}^{n} \mathbb{1}\{g_i = t\}, \quad (t = 0, \cdots, m-1),$$

  samples from the t-th generation, where $\{g_i\}_{i \in [n]}$ are independent and uniformly distributed on $\{0, 1, \ldots m-1\}$ indexing previous-generation models. This setting describes the scenario where data generated by all past models are mixed in a pool from which the training data for the next generation is collected.

Experiments for these two cases are shown in Figure 6. Column 1 corresponds to *Fully Synthetic*, columns 2 and 3 for *Most Recent Models*, and column 4 shows the case of *Randomly Sampled Data*. As we can see, when the window size K is increased, total collapse is delayed but still eventually happens. Observe that there are multiple visible horizontal yellow lines in the second row for window size 1. This is because $\boldsymbol{p}^{(m)}$ can converge to any of the Dirac

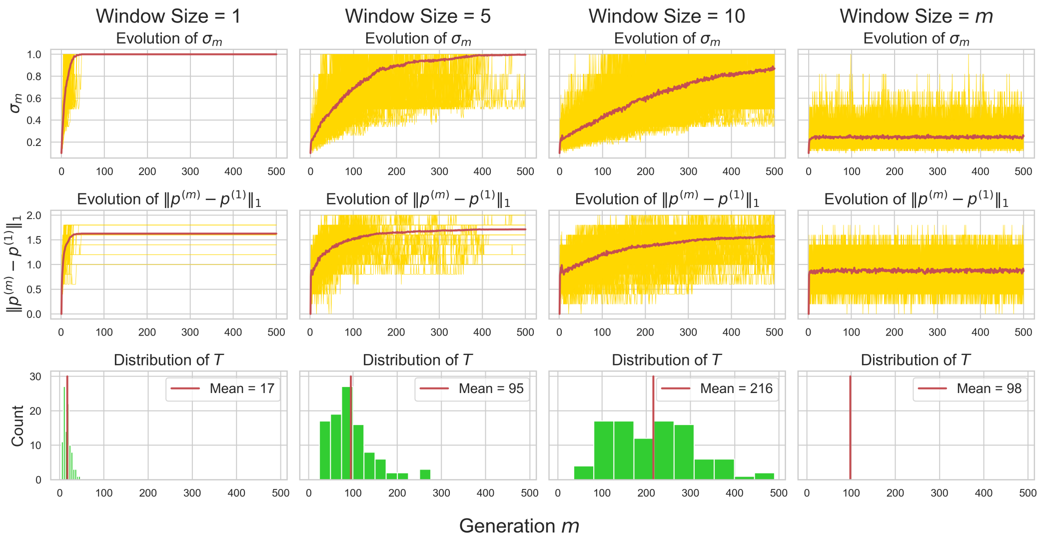

Figure 6: **Experiment for *Most Recent Models* and *Randomly Sampled Data*.** A hundred experiments were run for 500 generations for different window sizes and a fixed sample size $n = 10$. In the first two rows, each yellow line represents the evolution of σ_m and $\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1$ in one experiment respectively, with the red line being the empirical mean across 100 runs. The bottom row plots the histogram of total collapse time T with the red line being the empirical mean. The initial distribution $\boldsymbol{p}$ satisfies $s = 600$, $\tilde{s} = 52$ and $S_0 = 0.1$.

masses δ_i provided $p_i > 0$ by Proposition 1. A similar phenomenon can be observed in the second and third columns. On the other hand, in column 4 the model does deteriorate but the deviation plateaus very quickly. In particular, the 100 runs produce only one Dirac mass over the span of 500 generations, suggesting that randomly sampling from all the past models is qualitatively different from sampling from the most recent K models with a fixed K. We remark that for $K > 1$, even if $p^{(m)}$ is some Dirac mass, $p^{(m+1)}$ could still regain randomness from models before generation m, but with a fixed window size all the randomness eventually disappears.

5 Discussions & Conclusion

In this paper, we studied model collapse in language models through a simple statistical model. We provided theoretical analysis when training with only synthetic data and when adding real data from the original distribution. Our results demonstrate that model collapse always happens when the model is training solely on synthetic data, whereas controlling deviation from the initial distribution requires careful choice of the amount of synthetic data to inject in the training set. We also provided experiments showing that these findings extend beyond the simple theoretical settings.

Our current results describe only the statistical approximation error since all generation models are unbiased in our theoretical framework. However, as we discussed in the previous section, this framework can be extended to account for the functional approximation error by considering high-dimension Gaussian vectors as context embeddings instead of canonical vectors. Another possible extension is to study the effect of in-context learning (Wu et al., 2023) on model collapse, which is a key feature of transformer-based models.

Despite the simple setting of our current investigation, we believe that it lays the groundwork for better understanding and mitigation of model collapse in language models, thereby opening the way for the development of more general theoretical frameworks to study next-generation language models dynamics.

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